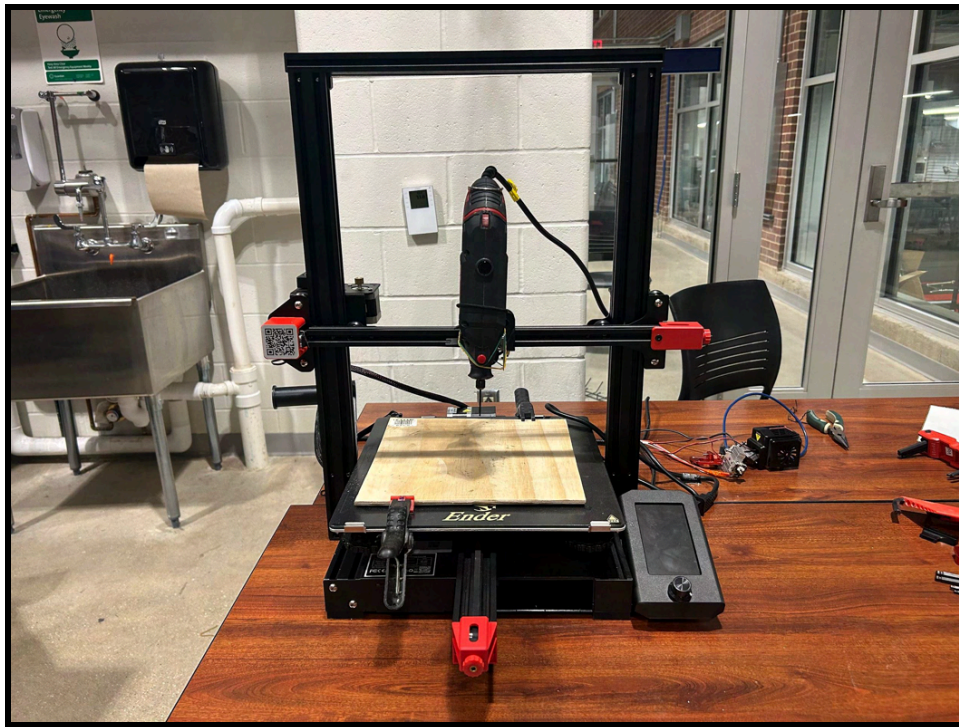


Project 3: Repurposing, Reuse, Recycling

ME 270: Design for Manufacturability

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ABA

1. ANALYSIS [80 points]

1.1.

a. Steel paper clips versus plastic paper clips;

Steel paper clips are made using galvanized steel wire, which is steel coated in zinc to prevent rusting when exposed to the environment. This steel is sourced from iron ore, which is blasted from mountains, crushed, refined, and then processed into pellets to be sent to a steel mill (UKEssays).

The galvanized wire is then wound into a small spool, which workers load into a metal stamping machine that bends the wire into the classic paperclip shape. The machine trims the paperclip to the standard size and sends it through three small wheels, which rotate the wire 180 degrees, then form the second and final bends. This process is extremely fast, and quality control doesn't add much cost since no complex testing is required to produce steel paperclips (UKEssays).

Plastic paper clips are typically made from polystyrene, a lightweight plastic that is both strong and flexible. The plastic begins as tiny pellets, which are melted at high temperatures. These are processed through a standard injection molding machine, which melts and injects the material at high pressure into a mold (Swaroop).

A carefully machined mold, usually made of mild steel, helps in this production (Swaroop). Injection molding is highly efficient, though the cooling phase can be tricky — the layout of the cooling channels and their interaction with mold pressure and cavity cooling is crucial for successful part production.

Repurposing injection-molded plastic paper clips is similar to repurposing other objects made from polystyrene. To reuse them beyond their typical function as clips, the material must be chemically broken down. Since polystyrene is also found in Styrofoam, it can be reused in many forms. When treated with D-Limonene, a compound found in orange peels, polystyrene dissolves into a sticky substance with adhesive qualities like superglue. With proper safety measures and ventilation, this is a simple and effective method to repurpose plastic paper clips (Shiino).

b. Forged vs. cast crankshafts;

There are three primary methods for manufacturing crankshafts: casting, forging, and billet machining. The forging process starts by placing a heated chunk of billet into a series of presses, shaping it into the traditional crankshaft form. Machining follows forging to further refine the part, which involves additional moving components and results in some material waste (Allen). Forged crankshafts have a more refined and uniform grain structure and are generally stronger than cast crankshafts. Although both types have comparable tensile strengths, forged cranks typically have an elongation rating about three times higher than cast ones. Elongation rating refers to the amount a part stretches before breaking, expressed as a percentage of its original length.

Forged crankshafts come in two types: twisted and non-twisted, referring to how they are manufactured. Twisted cranks are forged as a single unit and then twisted into shape using large automated equipment. Non-twisted cranks are set into the proper orientation from the start, which results in longer tooling times but creates a better uninterrupted grain structure (Boechtel).

For context, a crankshaft is a crucial component in an internal combustion engine, responsible for converting the pistons' up-and-down movement into rotational motion that powers the wheels. Crankshafts endure immense stress, sometimes exceeding 19,000 pounds per cylinder (Boechtel).

Cast crankshafts are made using standard casting methods, where molten metal is poured into a mold to form the rough part. This process requires less finishing and machining than forging and can still reach tensile strengths of around 70,000–80,000 psi. The recommended horsepower limit for cast crankshafts is typically around 450–500 hp (Allen). The process for manufacturing cast steel crankshafts is similar to that of cast pistons, and it's common in aftermarket parts. Cast iron, steel, and nodular iron are the most commonly used casting materials. The parting line, which is the seam where mold pieces meet, varies significantly in size (Allen).

Crankshafts are extremely heavy, which can be both an advantage and a drawback for repurposing. A forged crankshaft usually weighs more than a cast one, averaging 50–65 pounds. Traditional gym weights are often made of cast iron, which can be forged at home by amateur blacksmiths. Repurposing a forged crankshaft into several dumbbell weights cast in sand at home is a practical and efficient way to reuse an otherwise discarded car component (Nguyen).

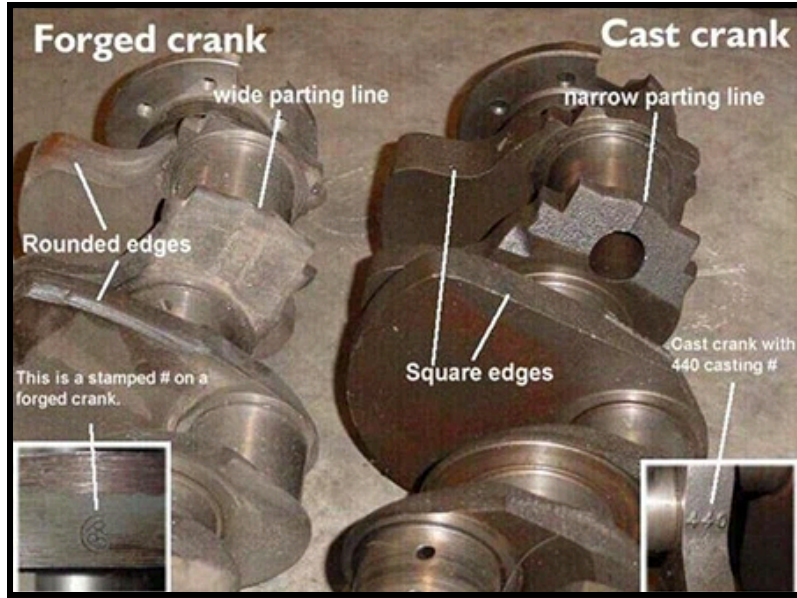


Figure 1. Forged vs. Cast Crankshafts

c. Plastic vs. glass pitchers;

To approximate plastic pitchers and how they're manufactured, examining the production of milk jugs gives similar insights and functions to a traditional pitcher. Milk jugs are made using high-density polyethylene (HDPE), which is blow-molded in large quantities. HDPE is a type of resin that begins as pellets and is melted into a small plastic tube. Air is then blown into this tube, which causes the plastic to press against the sides of the mold, forming a thin wall around the jug (Kershner). However, this method lacks precision, and large pools of plastic often form near the mold edges, where clamps hold the two-part machined mold together. Trim knives are used to remove the excess plastic around the edges (Kershner).

The HDPE method for producing large jugs is slower than other manufacturing techniques, and there are several challenges due to inconsistencies in plastic flow. Cooling times and ideal conditions for air and plastic distribution — the “Goldilocks zones” — are difficult to fine-tune. The machines also run continuously, 24 hours a day. This process creates significant plastic waste and storing and packaging each unit individually adds to the cost (Kershner).

Glass container production typically involves three main stages: handling and processing raw materials, forming and molding the containers, and finally, packaging and inspecting the final products. Raw materials like silica and soda lime are used to lower the melting point of sand and to make the glass more durable and glossy (Ajanta).

The raw materials are first ground into a fine powder using a grinding machine. This powder is then weighed and combined into the correct proportions for forming pitchers, creating a mixture called frit. The frit is melted in a furnace at around 1748 degrees Celsius. Molten glass is then directed through channels, cut into smaller pieces, and blown into the desired shape. Afterward, the pieces are heat-treated and cooled in a highly controlled process on a moving conveyor, known as annealing (Ajanta).

Cullet, which refers to discarded glass bottles or broken glass fragments, can also be used. Once glass pitchers made from cullet are broken or no longer in use, the glass can be added to cement composites. These composites have shown improved density, water absorption, and compressive strength, and they retain these qualities at higher temperatures. Although this repurposing method is highly niche and used for specific applications, it makes good use of material that would otherwise go to waste (Poweżka).

d. Paper vs. ceramic cups/mugs;

Paper cups are mainly made from paperboard, which originates from wood chips that are turned into pulp. This pulp is then cut into circular shapes and sent through a cup-forming machine to produce standardized cup blanks. Heat and pressure are essential during this process to ensure that the cups are sealed properly, both for safety and insulation purposes (Numrich).

More specifically, die-cutting machines are used to perforate each paper sheet that becomes a cup, while bottom knurling machines are responsible for sealing the cup bases to prevent leaks. The cup's bottom and body—known as blanks—are manufactured with strict tolerances and precise design standards (KETEGroup). Paper cups lined with polyethylene take approximately 20 years to degrade and can be produced at a rate of about 330 per minute. While highly economical, they present long-term environmental concerns.

Ceramic cups are created in two main ways: by using pottery wheels or through slip casting. Low- or high-temperature clay is selected, cut to size, and shaped using a pottery wheel in the first method, which is slow and done one piece at a time. After shaping, the cup is fired in a kiln to harden and treat the clay. Earthenware is the most common ceramic used for everyday cups and is fired between 1553–1700 K (Knoji).

Slip casting, on the other hand, involves mixing clay powder—often porcelain or earthenware—with a liquid dispersant to form a slurry, which is poured into a plaster mold. Once partially dried and drained, the mold's outlet is plugged, and the mold is split at its seams to remove the cup. It is then placed in a drying oven and coated with a glaze that gives it a shiny surface (Knoji).

Composting is a viable method for repurposing paper cups depending on the contents they've held. Most compost needs some dry, biodegradable matter, and paper cups without a plastic or wax lining are great additions to compost, helping balance its moisture level. However, only specific types of paper cups are compostable, as many have coatings that don't break down. If it's clear that a cup is made solely of paper without a liner, it can significantly enhance compost quality (Abeysekera).

e. Sheet-metal vs. cast hubcaps;

Progressive die stamping is the primary method used to produce sheet-metal hubcaps. Strips of sheet metal are gradually fed into a press, unrolled from a large coil, and guided into the die, which perforates the metal in a specific pattern. One major benefit of progressive die stamping is its rapid production speed, allowing for high volumes of hubcaps to be manufactured efficiently. Additionally, scrap metal produced from this process results in less waste compared to traditional machining methods. Each stamped hubcap is consistent and reliable, thanks to high-precision tooling and tight tolerances (Lampton). Each hubcap takes approximately 5–10 seconds to stamp, and requires minimal post-processing (Lampton).

The alloy typically used in casting hubcaps is a blend of aluminum with smaller amounts of silicon, copper, and magnesium to form a strong aluminum alloy. A design is first modeled, and a steel mold is machined to handle the high heat and pressure of the casting process. The molten aluminum alloy is then injected into the mold at high pressure to achieve both surface quality and structural strength. Die casting a hubcap takes about 2–5 minutes, with additional time required for cooling, trimming,

and polishing, which slows down the automation process (Easiahome).

Scrap aluminum sheet metal from the progressive die stamping process, as well as aluminum from hubcaps, can be repurposed in various ways. Aluminum hubcaps, for instance, can be transformed into serving trays or platters as a form of home decor. Repurposing them in this way involves applying a food-safe paint or varnish, a detailed process used to sanitize the surface. Given the durable aluminum structure of hubcaps, they can be made dishwasher-safe after finishing and are highly reliable for serving food (McMorrow-Hernandez).

f. Steel vs. copper nails

Much like the production of stainless steel paperclips, steel nails are typically made from spools of metal wire. This wire is fed into a multi-part nail-making machine. A gripper die secures the steel wire coil, while a large mechanical hammer strikes the wire, forcing it into a standard die to form the nail head. Shaped cutters then slice the wire on the opposite end of the nail head, sharpening it into a point and trimming any excess (Gale).

Next, the nails are passed through a rotating barrel filled with caustic soda, which cleans off grime left from the nail-forming process and removes small bits of metal. After cleaning, the nails are magnetized so they line up neatly in packaging. Post-processing for steel nails generally includes some sort of coating—such as zinc, phosphate, or paint—to protect them (Gale).

Copper nails come in several types, each designed for a different use: solid copper nails, threaded versions, and round-head styles. The process for producing copper nails is quite similar to that of steel nails, using copper wire of varying diameters. In construction, they are valued for their resistance to corrosion, especially in applications like copper roofing. The copper used is an alloy containing small amounts of zinc, tin, and aluminum. Additionally, copper nails conduct electricity and have antimicrobial properties. Due to copper's softness, less energy and tool wear is involved in manufacturing. Unlike steel nails, copper nails don't need heat treatment (Chen).

Because of their pure copper content, these nails can also be used to kill tree stumps over time. Copper alters the pH level of tree sap, slowly weakening the stump from the inside. This method can be useful in managing invasive trees in native woodland areas. It also offers a way to reuse damaged

copper nails, like those with stripped threads or bent shanks. The copper disrupts the tree's ability to transport water, nutrients, and enzymes as it gradually leaches into the system (Trout Brook Arborists).

1.2.

A circular economy works to keep materials and products in use for as long as they remain useful. It depends on systems that promote reuse, repurposing, and recycling—such as regeneration processes, flow management, and sustainable business models (EPA).

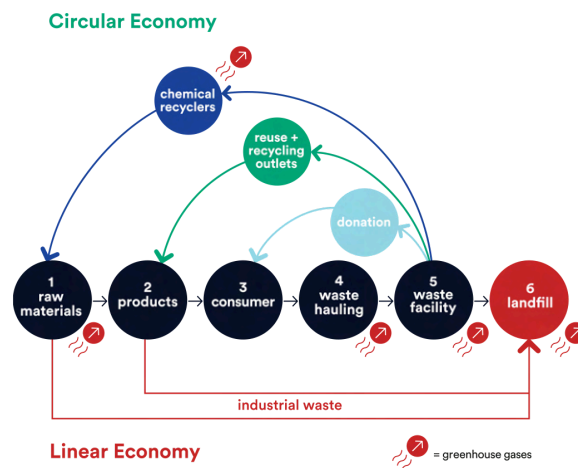


Figure 2. Circular Economy Graphic

The U.S. Environmental Protection Agency supports circular economies as part of its sustainable materials management efforts. This method helps reduce negative lifecycle effects, including those that impact the global climate. Recycling plays a major role in a circular economy, but it isn't the only focus. Products can be refurbished, resold, or designed with their end-of-life in mind, making them easier to repair and recycle. Circular economies are different from linear models, which typically follow a product → consumer → landfill path, as shown above. Strengthening recycling systems, increasing donation opportunities, and fostering a culture that supports reused goods all help support circular economies (RTS).

1.3.

Cradle-to-cradle is one design approach within the Life Cycle Assessment (LCA) framework, which evaluates the environmental impacts of a product throughout its life. This method emphasizes the potential for a product to be upcycled or refurbished after its initial use—essentially giving it a second life. In terms of product design, cradle-to-cradle focuses on strategies for end-of-life, such as reusing components in new products or recycling materials at the resource level to create entirely new goods. This approach brings environmental benefits, reduces costs by reusing materials, and results in a smaller overall ecological footprint (Nickel).

In contrast, the cradle-to-grave model tracks a product's full life cycle from raw material extraction (the “cradle”) to final disposal (the “grave”). It follows the traditional linear economy model and includes five main stages: raw material extraction, manufacturing, transportation, use, and disposal (Nickel).

While both cradle-to-cradle and cradle-to-grave examine a product's complete life span, cradle-to-cradle provides sustainable alternatives for reusing materials, whereas cradle-to-grave often leads to greater environmental harm, higher costs in raw material sourcing, and increased ecological damage (Nickel).

1.4.

Four disposable products fit for analysis are disposable water bottles, Ziplock bags, standard batteries, and disposable toothbrushes. Reusable products include stainless steel water bottles, Tupperware containers, reusable batteries, and reusable toothbrushes.

Disposable Water Bottles	
Materials Used	Polyethylene terephthalate (PET) plastic body, polypropylene caps, polyethylene labels
Material Selection Process	Lightweight, inexpensive, food-safe plastic that's recyclable.
Product Manufacturing	Injection-molded PET pellets are reheated and blow-molded into bottles, injection-molded water bottle caps.
Manufacturing Selection Process	Blow-molding is incredibly cost-effective and fast – producing thousands of bottles an hour.

Source: BPF

Stainless Steel Water Bottles	
Materials Used	304 Stainless Steel, silicone gasket, polypropylene cap, powder-coated exterior.
Material Selection Process	Stainless steel is food-safe, corrosion-resistant, and a good insulator; silicone provides ample seals and is inexpensive; all products are durable and reusable over sustained time periods.
Product Manufacturing	Steel deep drawing and spinning, trimming and annealing processes, welding the base, powder coating for refinement, and eventual assembly.
Manufacturing Selection Process	Steel deep drawing and spinning effectively forms cylindrical shapes with insignificant amounts of welding; powder coating is customizable and durable.

Source: Everich

Disposable Ziplock Bags	
Materials Used	Low-density polyethylene (LDPE), linear low-density polyethylene (LLDPE), colorants
Material Selection Process	Flexible, durable, moisture-resistant, heat-sealable plastics.
Product Manufacturing	Plastic film extrusion through a slot die, heat sealing to create the bag sides, cutting and packaging.
Manufacturing Selection Process	Plastic extrusion is continuous and inexpensive, and heat sealing ensures the product is airtight at high speeds.

Source: Trung Hoan

Tupperware Containers	
Materials Used	Polypropylene tubs, TPE seals
Material Selection Process	Polypropylene is food-safe and food-grade, while being durable, microwave-safe, and reusable.
Product Manufacturing	Injection molded-plastic for tupperware base and lid.
Manufacturing Selection Process	High-volume repeatable production with decent tolerancing; incredibly scalable and allows integration of snap-fit plastic mechanisms and flexible sealing measures.

Source: Inline Plastics

Alkaline Batteries	
Materials Used	Steel shell, zinc powder, manganese dioxide, plastic insulators
Material Selection Process	The cathode and anode and their reactive qualities power the battery, as well as a steel shell to protect from battery rupture and conduct electricity.

Product Manufacturing	Metal stamping to form the steel casing; electrode insertion between the cathode and anode, sealing the negative terminal for proper usage, packaging.
Manufacturing Selection Process	The majority of these parts are incredibly automated, so production can be easily scaled. Dry cell chemistry to compose these batteries is cost-efficient and tested.

Source: Microcell

Reusable Batteries	
Materials Used	Nickel, cobalt, lithium, manganese for battery composition, plastic casing, copper and aluminum, electrolytes (LiPF ₆).
Material Selection Process	Recharging properties without major chemical degradation, sealant processes that are safe and durable, higher cycle life and energy density per battery.
Product Manufacturing	Electrode coating on the internal shell; full cell assembly; sealing and charging processes; casing and externally labeling the batteries/
Manufacturing Selection Process	Automated assemblies allow for quick composition of an entire battery and safe means to combine reactive elements at repeatable measurements.

Source: ONREL Energy Storage Research

Disposable Toothbrushes	
Materials Used	Polypropylene, polystyrene, Nylon, plastic blister packaging
Material Selection Process	Plastics are moldable, cheap, and can withstand short-term use. Nylon has properties that can effectively serve hygienic purposes.
Product Manufacturing	Injection-molded handle, bristle tufting of the Nylon, trimming and packaging to finalize the product
Manufacturing Selection Process	Injection molding provides repeatable results that can be mass-produced; nylon bristle tufting automates insertion of each fiber efficiently.

Source: Khatabook

Reusable Toothbrushes	
Materials Used	Stainless steel handle, bamboo handle, nylon bristles, silicone grip
Material Selection Process	Head replacements reduce overall material use, stainless steel is durable and withstands water and other elements, bamboo is biodegradable.
Product Manufacturing	CNC machining or carving for handle formation; plastic injection molding for the brush head, nylon bristle tufting.
Manufacturing Selection Process	CNC machining ensures repeatability and durability in parts, and modular design with replaceable heads allows for less waste in an entire product.

Source: Snow Oral Care

The main criteria and limitations when designing disposable products are low upfront cost, mass production capability, and minimal production time. Common plastics like low-density polyethylene are often chosen, along with manufacturing methods like injection molding. These techniques are favored because they support continuous, highly efficient production, delivering precise and repeatable results with relatively low tooling costs—ideal for large-scale output.

In contrast, reusable products typically involve slightly more expensive, durable materials. Stainless steel, corrosion-resistant alloys, and strong plastics are preferred because they can endure repeated use and exposure to various elements. These materials also degrade more slowly and are often suitable for refurbishing or upcycling. Although the initial cost to consumers is higher, reusable products offer extended lifespans and enhanced performance, making them cost-effective over time. Manufacturing reusable products usually requires more complex assembly and a combination of materials to optimize function. They're also more likely to offer customization options and are produced in lower volumes.

1.5.

a)

Given:

Mass (kg)

Steel - 0.20

Wrought Aluminum - 0.10

Copper - 0.12

PP - 0.70

Glass - 0.28

Price (secondary) (\$/kg)

Steel - 0.40 - 0.50

Wrought Aluminum - 1.6-1.7

Copper - 4.8-5.3

PP - 0.8 - 1.0

Glass - 1 - 1.2

Minimum potential recovery value:

$$\begin{aligned} &= (0.2 \times 0.4) + (0.1 \times 1.6) + (0.12 \times 4.8) + (0.7 \times 0.8) \\ &\quad + (0.28 \times 1.00) \\ &= \$1.56 \end{aligned}$$

Maximum potential recovery value:

$$\begin{aligned} &= (0.2 \times 0.5) + (0.1 \times 1.7) + (0.12 \times 5.3) + (0.7 \times 1) \\ &\quad + (0.28 \times 1.20) \\ &= 1.942 = \$1.94 \end{aligned}$$

Range:

\$1.66 - \$1.94

b)

If you were to set $P(t) = 0$, and $P(t) = \text{Value} - 35 \times (t/3600)$, where value = \$1.80, in between \$1.55-\$1.94.

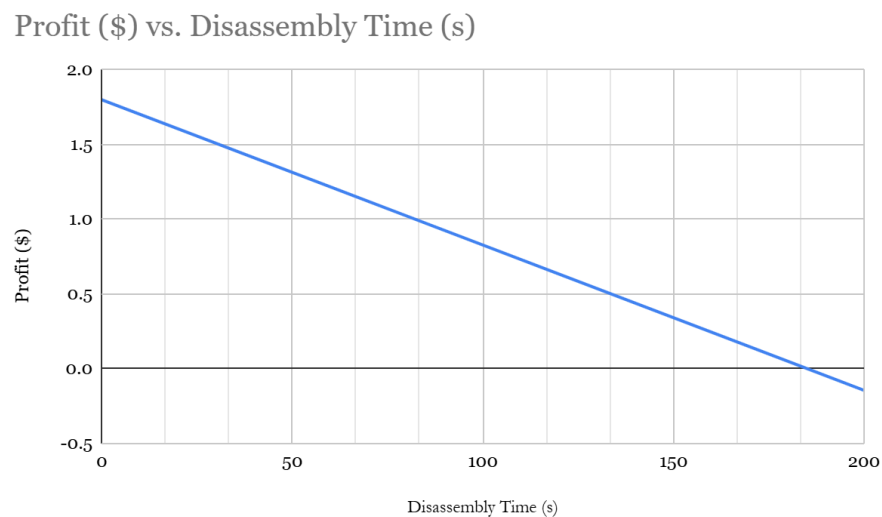


Figure 3. Graph of Profit vs Disassembly Time

The disassembly time break-even point occurs at approximately 190 seconds.

(c)

Material	Primary Energy	Secondary Energy	% Reduction
Steel	29-35	8-10	71-77%
Aluminum	200-215	18-19	91-92%
Copper	68-74	17-19	73-77%
PP	85-105	36-44	50-58%
Glass	24-26	11-12	50-54%

(d)

For each energy price, I used the equation

Profit = Material Value + Energy Rebate - Labor Cost.

$$= 1.80 + (\text{Energy Saved} * \text{Energy Price}) - (35/\text{hr} * \text{disassembly time (hours)})$$

Profit vs. Disassembly Time for Varying Energy Prices

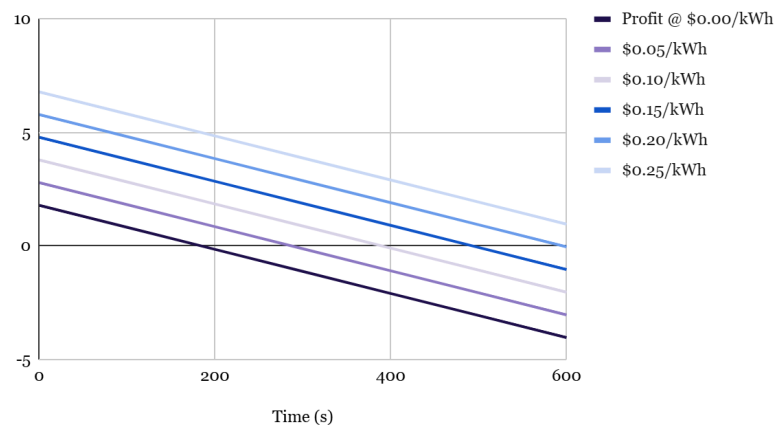


Figure 4. Profit vs Disassembly Time

2. DESIGN CHALLENGE [95 points]

2.1.



Figure 5. Ender 3 Pro

Over the last few weeks, I came into possession of a slightly damaged Creality Ender-3 Max 3D Printer. This is a standard desktop-sized 3D printer kit with a build volume of 11.8 x 11.8 x 13.4 inches. It has an all-metal frame and uses a linear pulley motion system that needs to be manually leveled. The printer can produce layers as thin as 100 microns, includes a 350W power supply, and weighs about 21 pounds (Gehrke).

In the foreground of the picture, there is also a small rotary tool. My design objective is to build a compact, desktop-sized modified CNC router that can carve small amounts of wood. By using the rotary tool's motor along with the three axes of the 3D printer, I hope to carve into thin sheets of wood.

The Ender 3 Max uses eight 2020/2040/2060 aluminum extrusions as its structural frame. These are manufactured through aluminum extrusion: an aluminum billet is heated in a furnace and pressed through a custom-fit steel die in an extrusion press. After extrusion, the sections are cut to size, stretched for straightness, sliced again, and finished in an aging oven to harden and stabilize the material (AEC).



Figure 6. 80/20 Aluminum

Approximately 20 M4 and M5 screws are required. These are produced using a cold-forming process, where a reciprocating die is fed a continuous spool of lubricated, heated metal wire. A series of dies punch out the screw head shape, and the screw blanks are then fed through thread-rolling dies to create the threads (ACCU).



Figure 7. M4 Screw Head

Two Z-axis T8 screw rods are used to move the tool head vertically. These rods are made of stainless steel, chosen for its corrosion resistance. The steel is first processed through a rolling mill and then fed through thread-rolling dies that form the threads directly onto the rod surface. A straightening process ensures the rods meet precision standards before being cut to length (YGM Tools).

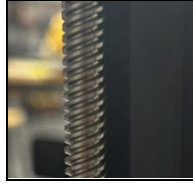


Figure 8. T8 Screw Rod

To link the motor and lead screw, two T8 lead screws and two shaft couplers are included. Shaft couplers are typically machined on CNC lathes, producing highly precise, durable parts. These components often include features such as a drain line, a coupling guard, and tight tolerances that ensure proper fit with other moving parts (Matthews).



Figure 9. Shaft Couplers

The motion system is powered by five NEMA 17 stepper motors. Each motor includes a stator made from silicon steel laminates shaped into a hollow cylinder, with coils wound around it to create the multiple stepping stages. The shaft is machined from stainless steel and bonded with magnets to form distinct poles. Bearings are installed at the shaft ends to allow smooth, controlled rotation (Lin Engineering).



Figure 10. NEMA Stepper Motors

Two GT2 timing belts are used to transfer motion. These belts are made from neoprene rubber reinforced with fiberglass cords. After being molded with trapezoidal teeth, the belts are cut to their

required length. The tooth profile ensures accurate meshing with GT2 pulley components (IPG).



Figure 11. GT2 Pulley Components

Four GT2 20-tooth nuts with 5mm bores interface with the belts. These are machined with very tight tolerances to ensure that the metric pitch between teeth matches that of the belt. Machining is favored for these parts due to the precision needed for smooth motion (IPG).

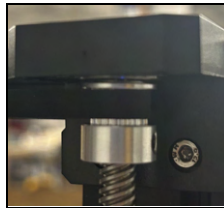


Figure 12. GT2 Tooth Nuts

A single 4010 DC fan provides cooling. The fan blades and housing are manufactured through injection molding. Inside the housing, a small DC motor drives the airflow. The fan's lightweight frame makes it suitable for use along the small GT2 track (3D Trade).



Figure 13. 4010 DC Fan

A 1.75mm filament tube made from PTFE is used for guiding material or wiring. This tube is produced using a paste extrusion process: fine PTFE powder is mixed with lubricant, compacted into a cylinder, and pushed through a small die to form the tubing (GREMCO).



Figure 14. PTFE Tube

A limit switch is used for position detection. Its enclosure is stamped from sheet metal, and the switch's electrical contacts are also formed through stamping. Internal springs are produced with a wire bending machine, and the switch is assembled with small, conductive internal components (Eaton).

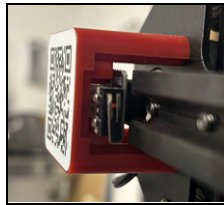


Figure 15. Limit Switch

The hotbed consists of an aluminum base plate that enables heat transfer, topped with a layer of tempered glass. The glass is manufactured through heat treatment or metal stamping. The hotbed assembly also includes gears and large screws that enable movement along the Z-axis (Tom).



Figure 16. Hotbed

Power is delivered via a 24V 15A power supply. This component is built on a small printed circuit board (PCB), populated with electronic components using Surface Mount Technology. A transformer is included in the circuit to manage voltage conversion and stability (PULS).



Figure 17. 24V Power Supply

Control of the system is handled by the Ender 3 4.1.1 motherboard. This board is manufactured from fiberglass and copper laminated with epoxy resin. Chemical etching is used to form the circuit pathways that enable communication with motors, sensors, and other peripherals (Qualitetch). (*Not pictured*)

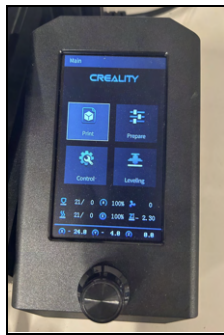


Figure 18. LCD Display

A small CNC desktop router can be purchased online for around \$800, semi-assembled and with a small working light, control board, and emergency stop button. There are similar dimensions to this 3D-printer, with a print bed of 15.75 x 15.75 inches.



Figure 19. CNC Desktop Router

Small aluminum profiles make up the base of this product, which are also fed through a similar extrusion process to the 80/20 or 40 series that is used in the Ender 3 Max. Twenty M5 screws are additionally utilized, whose manufacturing process is documented above.

An MDF spoil board slots into these aluminum extrusions. MDF, or medium-density fiberboard, is manufactured by breaking down wood into fine particles, combining them with a mixture of resin and wax, and pressing the mixture into panels at high temperature (Watkinson). This is the brown-colored flat board lined with holes in the image above.

An X-axis assembly sits on top of this half-assembled product, secured with several M5 bolts. A small, 3D-printed spindle holder is attached to this, where the spindle and its DC motor rest. This assembly provides the capability to carve into wood. DC motors are produced in a similar way to stepper motors — a large coil of wire wound around different poles, with a stainless steel machined shaft running through the part.

A large, pre-made controller is mounted to the side of the router, and screws easily into the side of the small bed. This controller contains a “control unit,” which houses a series of motors and driver components that move the machine’s axes. There is a wired connection between this controller and the motors powering the router, whether that’s through a set of motor encoders programmed with G-code or a pre-made stepper controller/PCB with pre-soldered components. This is made similarly to the PCB within the Ender 3 Max (AT Machining).

A large drag chain is what allows movement between the wired motors and their mounting locations on the aluminum extrusions. Unlike the rubber-toothed belts used in the Ender 3, a drag chain is more standard on full-sized CNC equipment. Steel rods are formed into pins and welded to side bars, which then undergo heat treatment for strength. A forming machine straightens and stamps this metal chain, and a pin is inserted and welded. Grinding processes polish the weld and close the links, followed by a round of testing and tolerancing to complete the process (Laclede).

Stepper motors for the X, Y, and Z directions are wired to sockets located at the ends of the aluminum extrusions, and the spindle for carving plugs directly into the controller socket. A router bit is screwed in, with precisely machined threads formed using the rolling thread-forming process.

The main challenge in repurposing the Ender 3 Max lies in the additional weight a rotary tool places on the 80/20 aluminum frame and the small neoprene tracks of the printer. The rotary tool available weighs approximately 2.5 pounds, which would place considerable strain on the printer's axes and stepper motors. NEMA 17 motors are known to burn out relatively quickly when subjected to excessive pressure.

Another concern is the difference in user interface and control systems. CNC routers often include clear and intuitive controls, along with direct software that is easy to follow, features that the Ender 3 lacks. For example, the presence of an E-stop button and a spindle directly connected to both a power source and controller makes it much easier to fully shut down or power on a CNC system, a convenience not present in the 3D printer.

However, the similarities between the two machines offer helpful advantages. Both utilize stepper motors across three axes of motion, making the transition between platforms feasible. Power supplies and motherboards are also comparable in composition and capable of interpreting various types of code, so adapting one system to another is within the realm of possibility.

Additionally, both machines share similar frame sizes and use large quantities of aluminum. This could help absorb the added strain introduced by the rotary tool. The inherent durability of aluminum, along with limit switches that prevent carving into the frame, act as critical safety features for both machines.

Economic and environmental factors were central to the decision to repurpose this printer. CNC routers and 3D printers generally serve personal or small-scale manufacturing purposes and aren't tied to large-scale global needs. As desktop-sized devices, both are often used in personal projects—something easily envisioned in a basement workshop setting.

From a cultural and social standpoint, there are minimal ties to either 3D printing or CNC routing. While both fall under the umbrella of manufacturing, cultural or social influences aren't major considerations here. On a larger scale, the manufacturing and international distribution of these machines may involve labor laws or tariffs, but those factors don't directly impact this project.

In terms of environmental impact, converting this printer into a CNC router is arguably more

sustainable than keeping it as a non-functional 3D printer. With the original nozzle no longer operational, the printer might otherwise become waste or require repair, only to continue extruding non-recyclable PLA or ABS filament. Wood carving, by contrast, can make use of scrap wood already available in a shop and avoids producing fumes or additional waste. Since the CNC router involves no heating process, it consumes less energy overall.

Economic concerns were perhaps the most significant contextual influence on this design. As is often the case in college life, funds are extremely limited. This redesign offers a cost-effective alternative use for a 3D printer while eliminating the recurring cost of filament. Wood carving has practical, everyday applications and introduces no ongoing expenses. The goal was to avoid large financial investments—and for others considering a similar path, this approach may offer a more accessible and affordable entry point into personal fabrication projects.

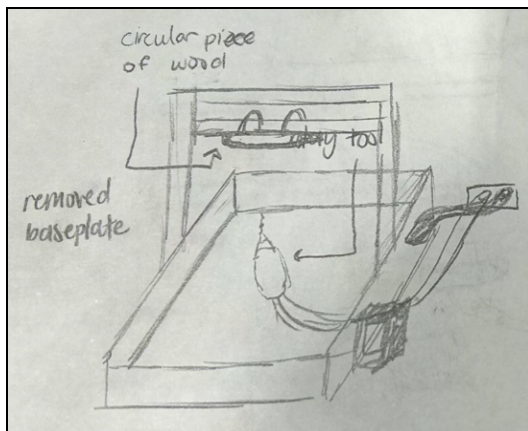


Figure 20. Sketch 1

A small dremel sits below the moving X-axis and carves wood directly.

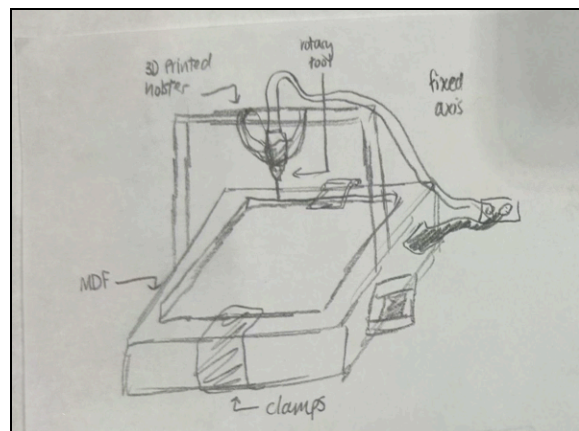


Figure 21. Sketch 2

80/20 Extrusions are cut from the top bed. They rest on a fix axis, where the rotary tool cuts and varies in string length.

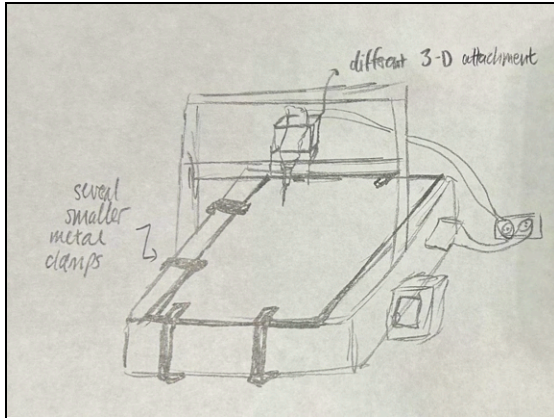


Figure 22. Sketch 3

Large metal clamps and a 3D attachment help the MDF and rotary tool fit in place.

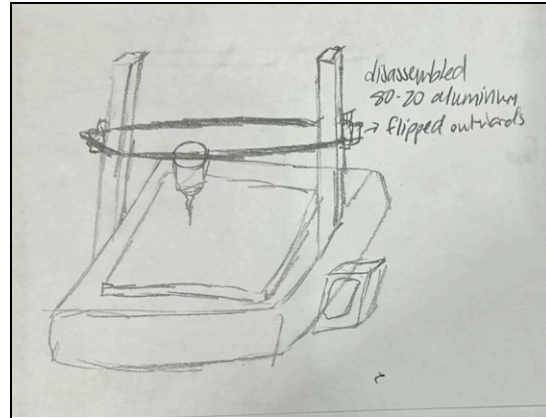


Figure 23. Sketch 4

Eliminating 80/20 extrusions as well as a lone belt tying the rotary tool define this feature.

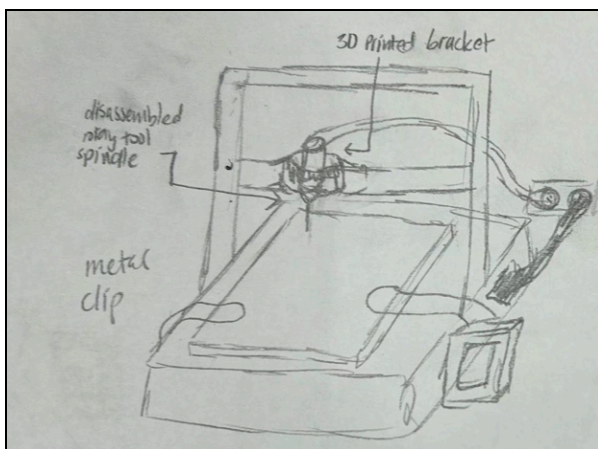


Figure 24. Sketch 5

The rotary tool is further disassembled into a more traditional CNC router spindle, allowing more freedom to rotate further on the X-axis



Figure 25. Sketch 6

MDF is held vertically and carved into with a vertically-located rotary tool.

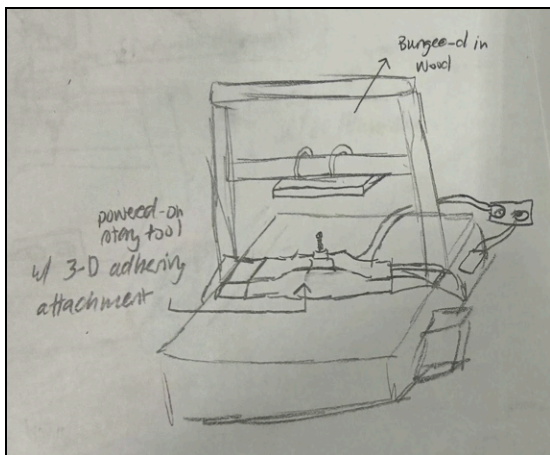


Figure 26. Sketch 7

Wood is adhered to the moving Z-axis part while the rotary tool is adhered to the hotbed with a 3D printed attachment.

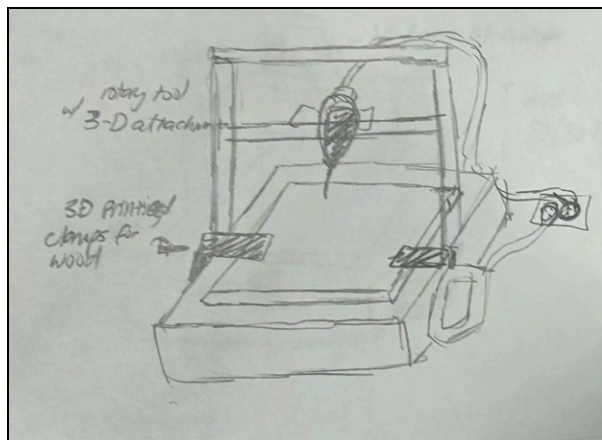


Figure 27. Sketch 8

3D Printed wood clamps adhere the MDF in place, and a traditional rotary tool with a small attachment adheres it in place with the belt and bearings.

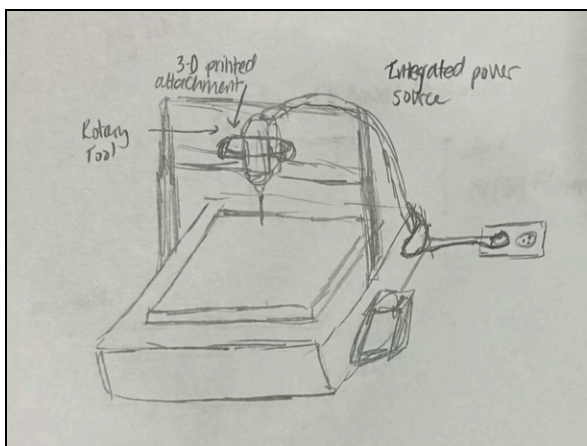


Figure 28. Sketch 9

The power source to the rotary tool and Ender 3 is integrated, as well as a traditional attachment and MDF placement.

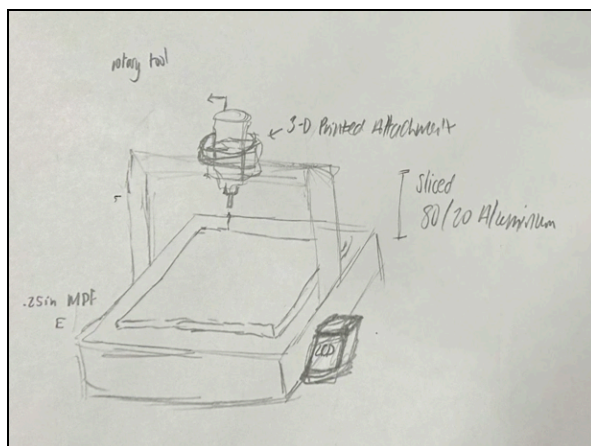


Figure 29. Sketch 10

80/20 aluminum is sheared to half the size in its z-axis, and a traditional 3D printed attachment is produced.

2.2.

Ender 3 Max Bill of Materials		
Part and Quantity		Repurposing/Recycling/Reuse
2020/40/60 Aluminum Extrusions	8	Repurposing
M4, M5 Screws	20	Repurposing
Z-Axis T8 Screw rod	2	Repurposing
T8 Lead Screw, Shaft Coupler	2	Repurposing
Nema 17 Stepper Motors	5	Repurposing
GT2 Timing Belt	2	Repurposing
GT2 20 teeth nuts, 5mm bore	4	Repurposing
4010 DC fan	1	Reuse
1.75mm Filament Tube	1	Reuse
Limit Switch	1	Repurposing
Hotbed	1	Repurposing
24V 15A Power Supply	1	Repurposing
Ender 3 4.1.1 motherboard	1	Repurposing
12864 LCD Screen	1	Repurposing

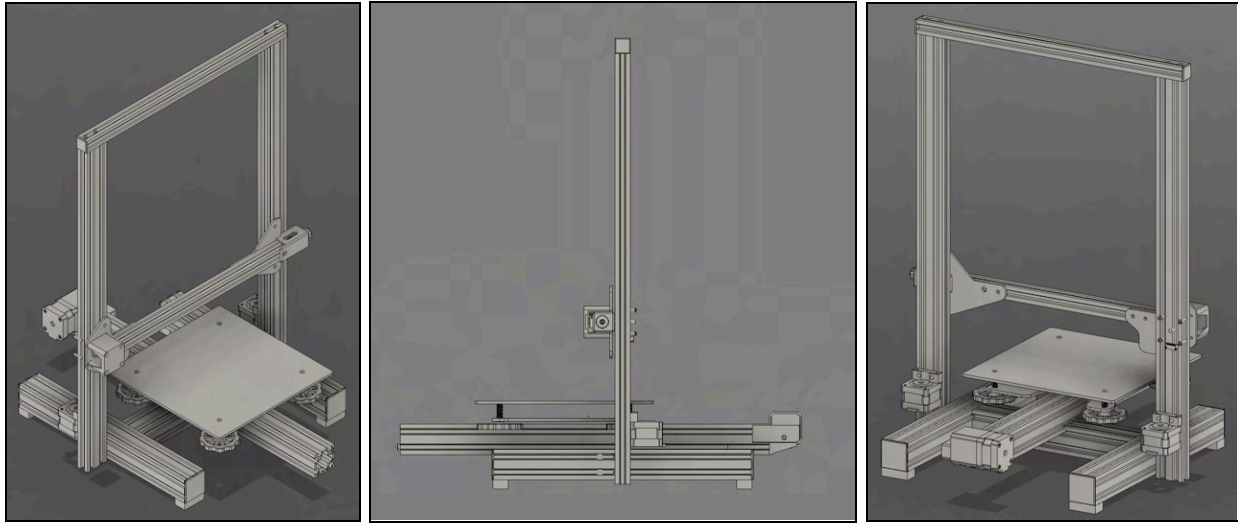


Figure 30. Several Views of Ender 3 CAD Models

2.3

This repurposed product would effectively carve wood using the three functional axes from a 3D printer. It could carve up to 0.25 inches into relatively soft wood and operate through G-code control. The intended end-user is an individual seeking personal use or someone interested in completing small freelance wood carving projects.

The Ender C(NC)

1. Performance

This machine's speed aligns closely with that of an Ender 3 Max—around 60 mm/s when extruding filament. However, since it's capable of moving up to 150 mm/s, wood carving may be faster. The anticipated precision is approximately 0.01 mm, on par with a typical 3D printing nozzle.

2. Environment

Designed for desktop use, this machine can endure standard gravitational forces in indoor environments. Adequate ventilation and dust management are essential during operation due to wood shavings and minor noise from the rotary tool. The device isn't expected to undergo intense physical stress, and its users are assumed to have a baseline understanding of manufacturing machinery.

3. Service Life

With consistent maintenance—such as replacing cutting bits and 3D-printed brackets—the product should last between 2 and 5 years. Strengthened 80/20 aluminum extrusions contribute to its durability and long-term feasibility.

4. Maintenance

Routine tasks like lubricating rails, clearing dust before each session, swapping out worn rotary bits, and inspecting belts will keep the machine running smoothly. Maintenance is straightforward, and offering replacement rotary bits as consumables could be beneficial for users.

5. Target Costs

After initial setup, few extra costs are involved. An Ender 3 Max typically costs around \$370, and an off-brand rotary tool around \$20. Beyond that, expenses include wood material and filament to print brackets. These costs are competitive, and the customizable, home-friendly nature of the product enhances its value.

6. Competition

Comparable desktop CNC routers range from \$300 to \$600 and often ship partially assembled. Marketing this as a DIY conversion kit for existing 3D printers allows for highly competitive pricing and adaptability.

7. Shipping

The total weight is about 15 pounds, with approximate dimensions of 20 x 20 x 20 inches. Shipping costs range from \$27 to \$32 depending on the carrier. Packaging must protect delicate components, and delivery would go directly from warehouse to customer through a contracted shipping service.

8. Product Volume

Mass production isn't the intent. However, if the modification process is optimized, partial automation and assembly could be introduced. Plastic components may eventually be injection

molded, and rotary tools might be stripped to essential elements for user assembly.

9. Packing

Packaging must securely house all individual tools, electronics, screws, and belts. The power supply should be insulated from heat and moisture. Shipping would cost about \$30 per unit—acceptable for a product of this caliber. Durable cardboard boxes and environmental protection are essential.

10. Manufacturing Facility

Given that most parts come from an existing 3D printer, current manufacturing plants could be adapted to house the additional conversion materials. No new facility is currently planned, but a dedicated assembly area with potential future expansion into a factory is ideal.

11. Size

The maximum size of this entire finished product would not extend past the size of an Ender 3 Max Pro, 20.2 x 22.17 x 23.23 inches. The minimum size would not deviate very far from this, as there is not much leeway unless large modifications involving slicing the aluminum extrusions are necessary.

12. Weight

The weight of this entire project would be around 25 pounds, factoring in the preexisting weight of the Ender 3 Max Pro and the approximately 2.5-pound rotary tool. 3-D printed attachment weight would be negligible, a few hundred grams at most.

13. Aesthetics and Finish

Appearance varies by user customization, but should retain a sleek profile aligned with the original printer. The rotary tool—based on a Hyper Tough model—and black ABS-printed brackets contribute to a clean, professional look. Wood clamps would also be printed in black ABS for consistency.

14. Materials

Materials include a functioning (or partially broken) 3D printer capable of 3-axis motion, a wood-cutting rotary tool, and PLA filament for clamps and attachments. Software that translates

G-code into CNC-readable instructions may also be needed.

15. Product Life Span

While the product can be supported for years, it will need updates as 3D printer designs evolve. As a novelty product, it suits hobbyists making small personal projects. Market growth and expanded use cases could drive further development.

16. Standards, Specifications, Legal Aspects

No strict national regulations apply to desktop CNC routers, though ANSI B11 safety standards should be observed. Due to risks posed by rotating tools, waivers may be needed. Clear warnings about safe operation are necessary.

17. Ergonomics

The machine should be entirely operable by hand, with custom-fit attachments for each component. Since 3D printers are already semi-assembled manually, users shouldn't face excessive ergonomic strain when modifying the unit.

18. Customer

The target audience is hobbyists or small-scale makers looking for affordable, precise, and high-quality wood carving. Ideal users are comfortable following manuals and have the skills to complete the conversion.

19. Quality and Reliability

There may be concerns about the strength of 80/20 extrusions and belt wear when bearing heavier tools. Calibration accuracy and axis precision are crucial. Defects in 3D-printed parts, such as weak layers, may also affect performance.

20. Shelf Life

The product is best stored in typical indoor environments. Its components won't degrade faster than a standard 3D printer. Since it uses an external power source instead of batteries, shelf life is not a

major concern.

21. Processes

Assembly instructions will be detailed and user-friendly. Manufacturing can rely on conventional methods, but packaging requires extra care to protect sensitive components.

22. Timescales

Assembly time varies with user experience. Most hardware can be printed, and basic tools and calibration parts are simple to fabricate. If kits are assembled on demand, delivery and setup times of under one week are achievable.

23. Testing

Testing includes strength checks for printed attachments, spindle stability, and woodcutting consistency. Additional evaluations should assess ease of installation and user adaptability across different printer models.

24. Safety

Users must be informed of the risks associated with high-speed cutting tools. Proper eye protection is essential. Safety labels must be included on the rotary tool and software to clarify that this is an aftermarket modification and not an officially supported printer upgrade.

25. Company Constraints

Constraints include printer compatibility, structural capacity of extrusions, belt durability, and overall rigidity. The product's precision also depends on how well users build and calibrate their kits.

26. Market Constraints

Initial distribution will be limited to the U.S. Domestic sourcing of rotary tools and components is preferred to reduce costs and maintain kit consistency. International markets may be considered later.

27. Patents, Literature, Product Data

Research on prior desktop CNC designs, DIY limitations, and safety concerns is necessary. Attention should be paid to existing patents on rotary tools and modification methods to avoid infringement.

28. Political and Social Implications

There are no significant political or social concerns. Market performance may be impacted by economic shifts, but the product itself is stable and unlikely to draw controversy.

29. Disposal

Disposing of the kit is complex due to multiple electronic parts, rechargeable batteries, and limit switches. Responsible disposal requires disassembly. Packaging can be made eco-friendly, and some filament types may be recyclable.

To effectively evaluate all the sketches above for an accurate Ender C(NC) design, I placed them into a comparison matrix to assess added cost for each iteration, feasibility, production and assembly time, and overall performance of the redesigned project.

Criteria	Sketch 1	Sketch 2	Sketch 3	Sketch 4	Sketch 5	Sketch 6	Sketch 7	Sketch 8	Sketch 9	Sketch 10
Added Cost	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1
Feasibility	-1	0	1	-1	1	-1	-1	1	1	-1
Time required	-1	-1	0	-1	1	-1	0	1	1	0
Aestheticness	0	0	1	0	0	-1	-1	1	1	0
Performance	1	-1	1	0	1	-1	-1	1	0	1
Total	-2	-3	2	-3	2	-5	-4	3	2	-1
Rank	5	6	2	6	2	8	7	1	2	4

Post-comparison matrix, Sketch 3 and Sketch 8 are the most highly ranked, effective products that could be manufactured.

The only difference between sketches 3 and 8 are the metal vs. small 3D clamps present in sketch 3/8. By incorporating both of these and utilizing plastic parts for clamps and rotary tool attachments, the end model will be more effective.

This is the CAD model I have presented of sketch 8.

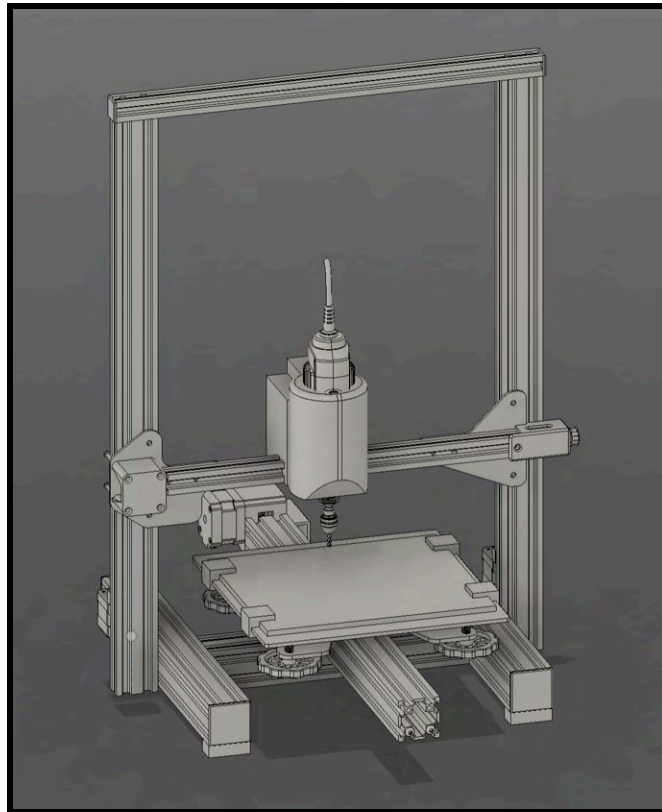


Figure 31. Sketch 8 CAD Model

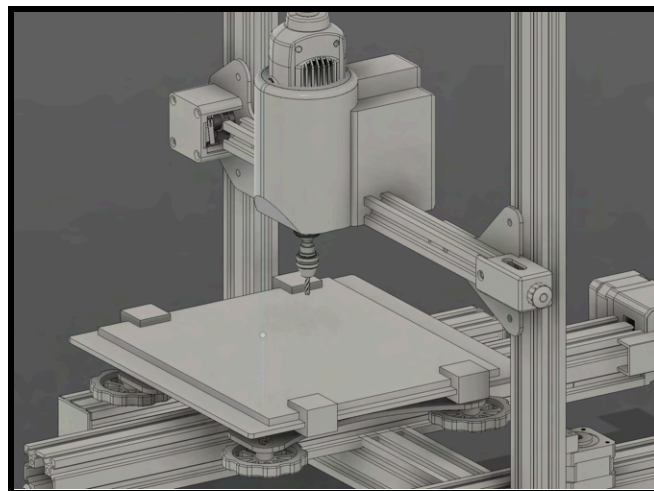


Figure 32. Zoomed-In Sketch 8 CAD Model

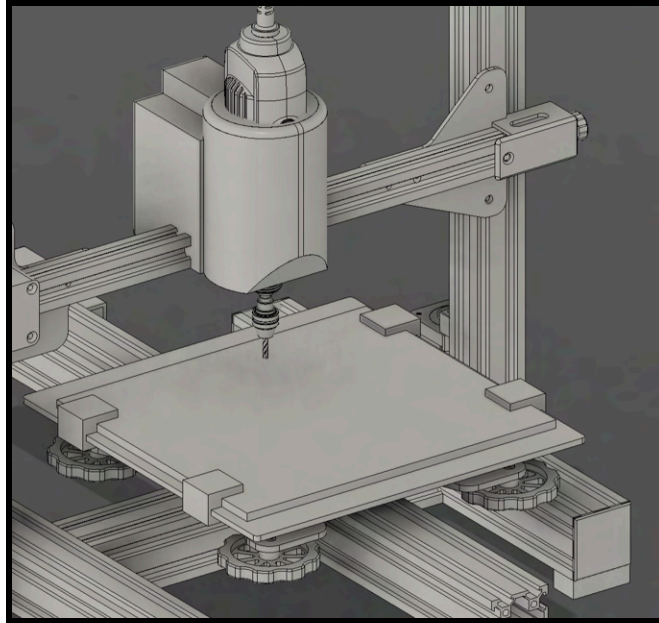


Figure 33. Sketch 8 View

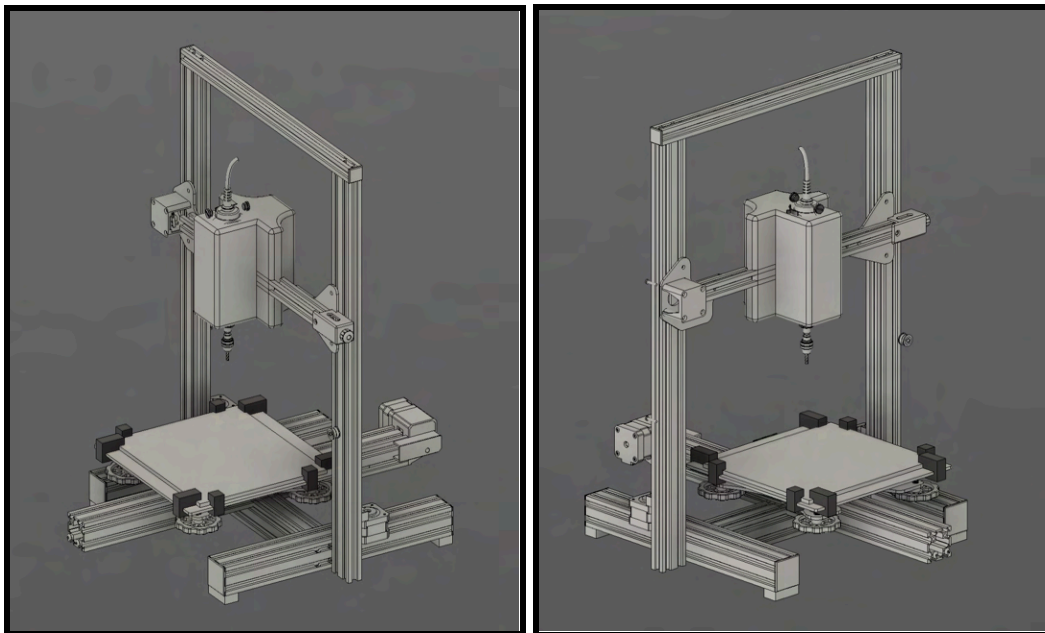


Figure 34. Sketch 3 View

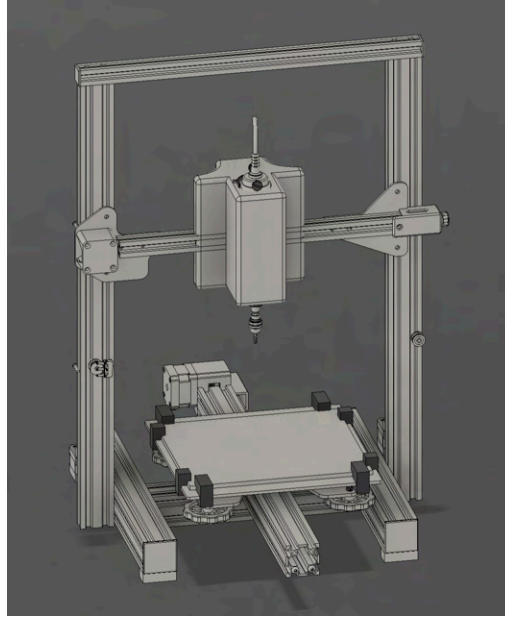


Figure 35. Sketch 3 View

For Sketch 3, I added a distinction between the small metal clamps that were present. To compare sketches 3 and 8, a modified Pugh matrix was used with five specific constraints: target costs, aesthetics and finish, customer appeal, timescales, and performance, to evaluate how well each model meets them.

Criteria	Sketch 3	Sketch 8
Target Costs	-1	-1
Aesthetics/Finish	1	0
Customer Appeal	0	0
Timescales	1	0
Performance	1	0
Total	2	-1
Rank	1	2

Based on the final PDS, the design selected for fabrication will be an iteration of Sketch 3. This version is the most effective in terms of performance, time efficiency, and aesthetics while still being

manufacturable. Before generating the engineering drawings, I made sure to fully update the CAD model to avoid producing ineffective drawings and to ensure they are as valuable as possible. While this technically means completing the steps out of order, I believe that having the most finalized and detailed design is essential for producing meaningful part drawings that best support the CNC's production.

This further updated CAD model is detailed and explicit, but lacks linkages. These will be the common M4 and M5 bolts expressed in the earlier BOM.

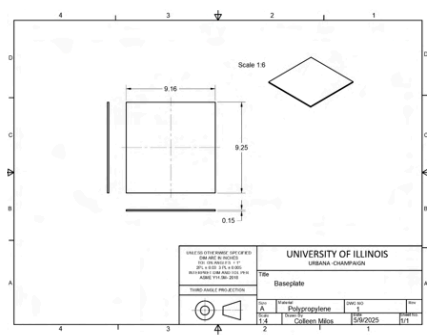


Figure 36. Ender 3 Baseplate

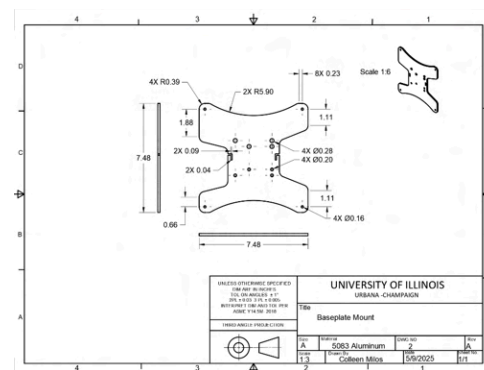


Figure 37. Baseplate Mount

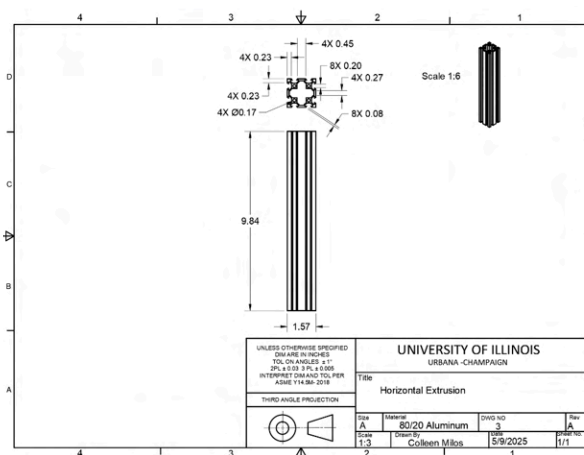


Figure 38. Horizontal Extrusion

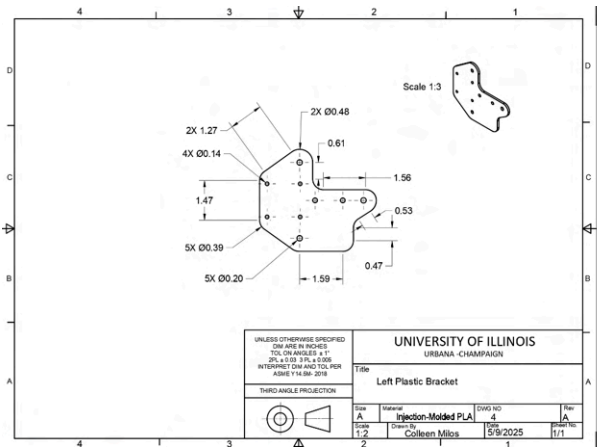


Figure 39. Left Plastic Bracket

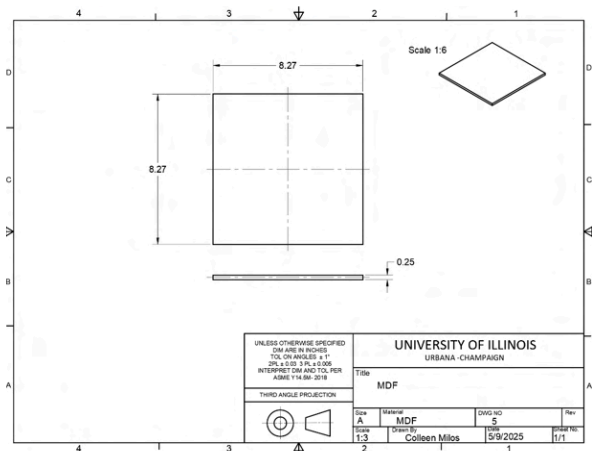


Figure 40. MDF

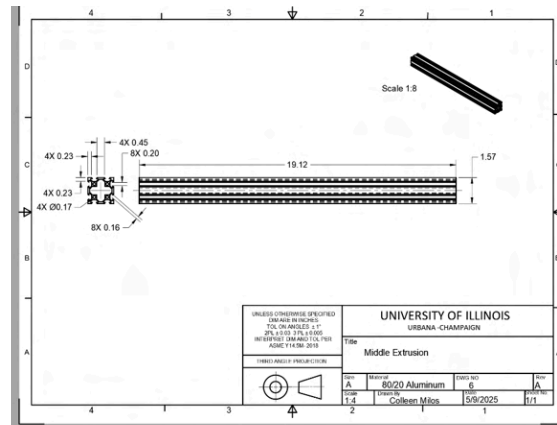


Figure 41. Middle Extrusion

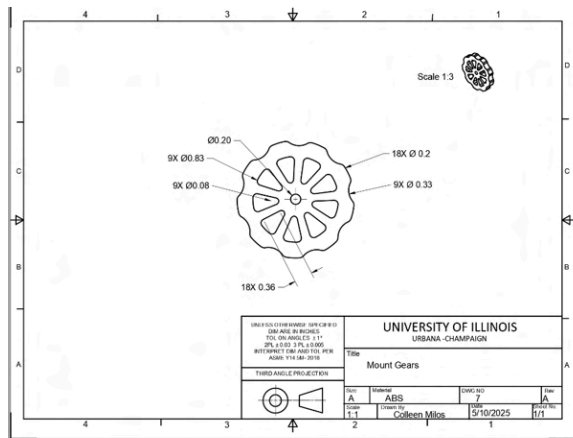


Figure 42. Mount Gears

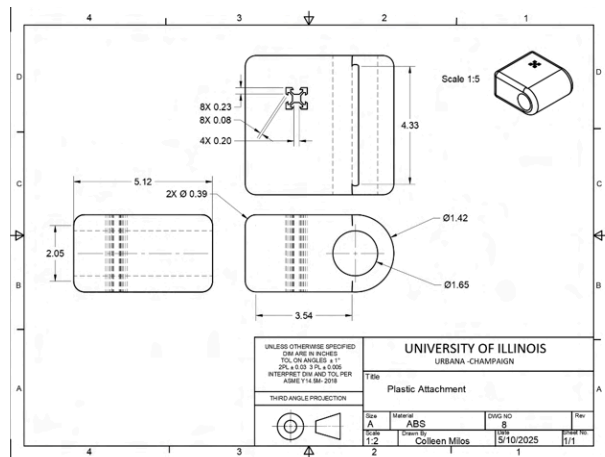


Figure 43. Plastic Attachment

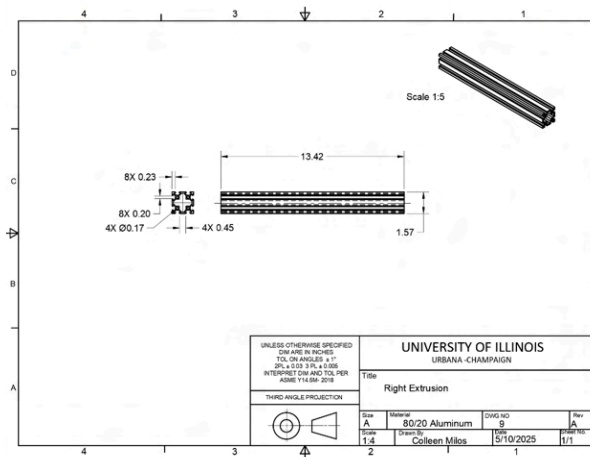


Figure 44. Right Extrusion

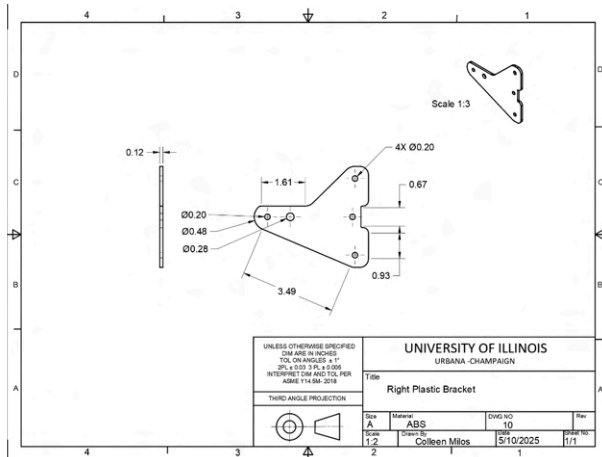


Figure 45. Right Plastic Bracket

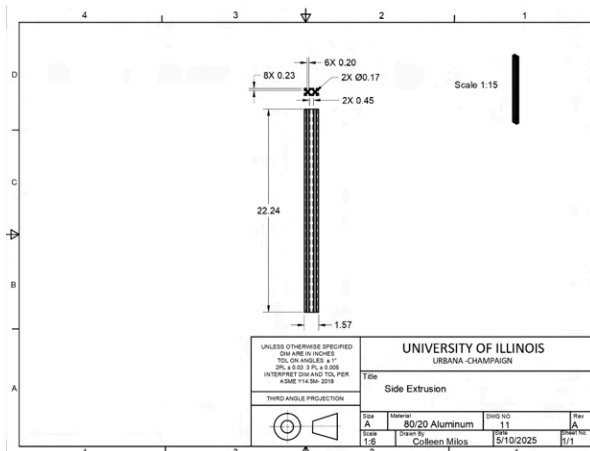


Figure 46. Side Extrusion

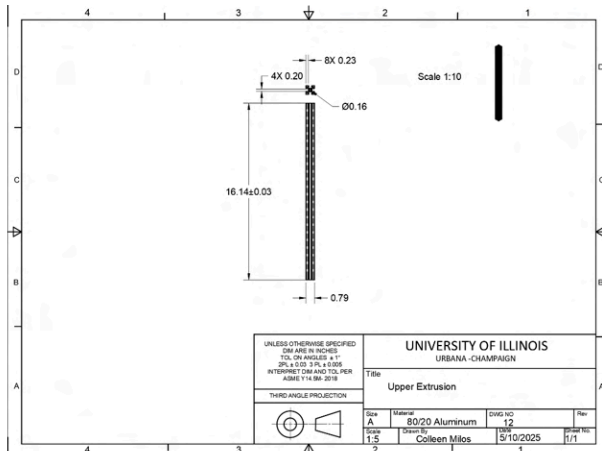


Figure 47. Upper Extrusion

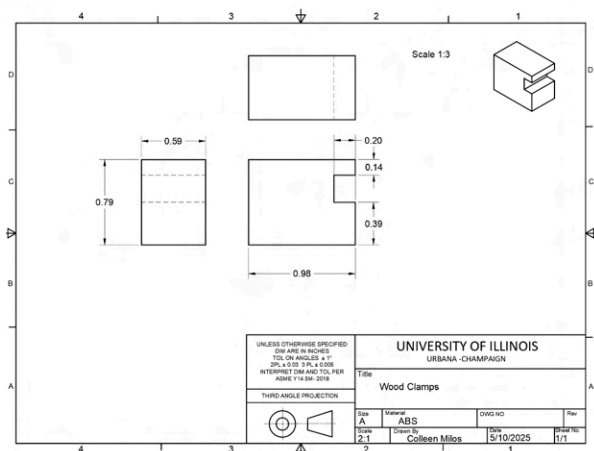


Figure 48. Wood Clamps

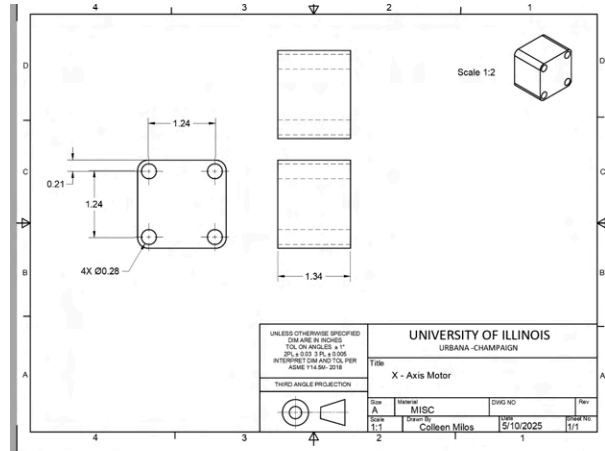


Figure 49. X-Axis Motor

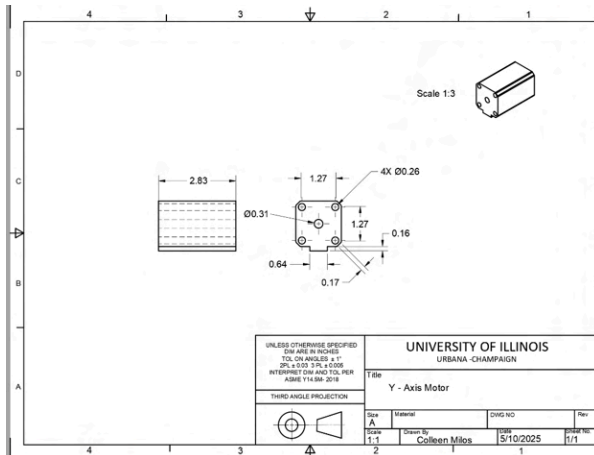


Figure 50. Y-Axis Motor

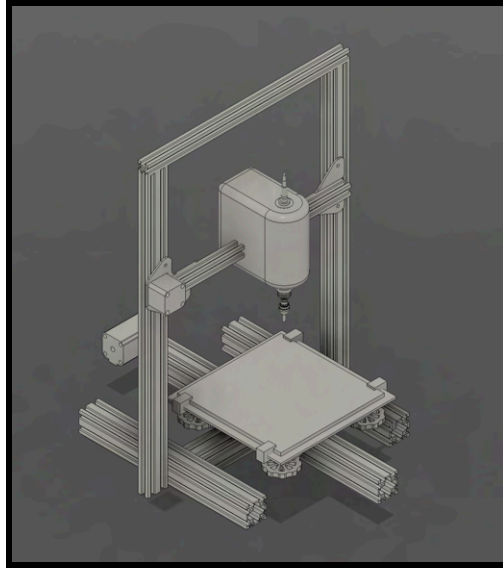


Figure 51. Updated CAD File

This is the CAD file after all possible revisions. Using Sketch 3, this is as complete of a work as possible. The plastic insert houses the dremel tool, and small wood clamps hold MDF in place.

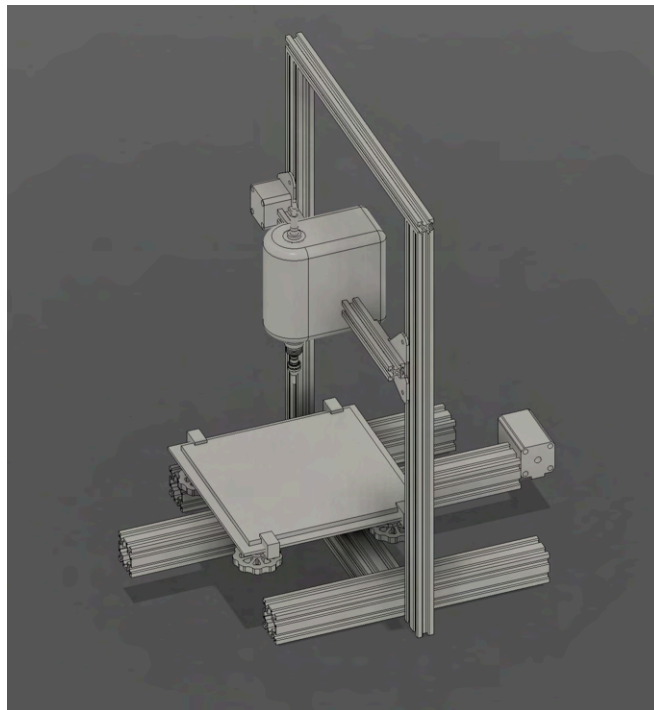


Figure 52. Alternate View.

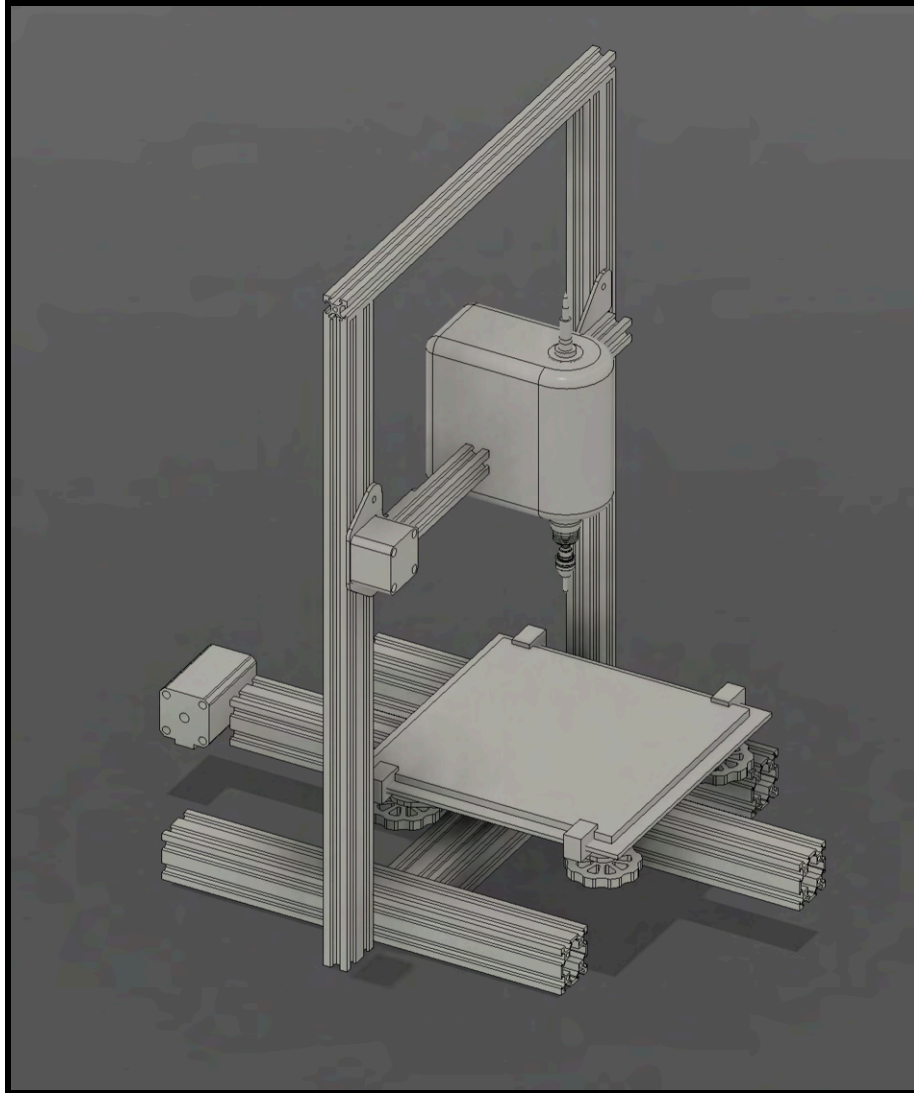


Figure 53. Alternate View

These are the GD&T referenced parts within my drawing. The measurement of 80/20 width is what was measured using inside caliper prongs, a micrometer, and outside caliper prongs.

Measurement 1			Measurement 2		
Inside Caliper			Micrometer		
	Measurement	Z-score		Measurement	Z-score
1	20.01	-0.2858050519	1	19.97	-0.6357704217
2	19.89	-1.335701161	2	20.11	0.5891083724
3	20.18	1.201547769	3	20.1	0.5016170299
4	19.97	-0.6357704217	4	19.96	-0.7232617641
5	20.18	1.201547769	5	20.21	1.464021797
6	20.23	1.639004482	6	19.89	-1.335701161
7	19.92	-1.073227134	7	19.97	-0.6357704217
8	20.14	0.8515823997	8	20.05	0.0641603177
9	20	-0.3732963944	9	20.03	-0.1108223671
10	20.03	-0.1108223671	10	19.92	-1.073227134
11	20.15	0.9390737421	11	20.16	1.026565085
12	19.9	-1.248209819	12	20.11	0.5891083724
13	19.9	-1.248209819	13	20.02	-0.1983137095
14	20.06	0.1516516602	14	20.01	-0.2858050519
15	20.08	0.3266343451	15	20.25	1.813987166
Average	20.04266667		Average	20.05066667	
SD	0.1142970232		SD	0.1047764059	
Min	19.89		Min	19.89	
Max	20.23		Max	20.25	
Confidence Interval	0.2285940465		Confidence Interval	0.2095528118	

Measurement 3		
Outside Caliper		
	Measurement	Z-score
1	20.19	1.289039112
2	19.89	-1.335701161
3	20.22	1.551513139
4	19.89	-1.335701161
5	19.98	-0.5482790792
6	19.92	-1.073227134
7	20.06	0.1516516602
8	20.04	-0.02333102465
9	20.18	1.201547769
10	19.89	-1.335701161
11	19.9	-1.248209819
12	20.25	1.813987166
13	19.89	-1.335701161
14	19.91	-1.160718476
15	20.15	0.9390737421
Average	20.024	
SD	0.1393249029	
Min	19.89	
Max	20.25	
Confidence Interval	0.2786498058	

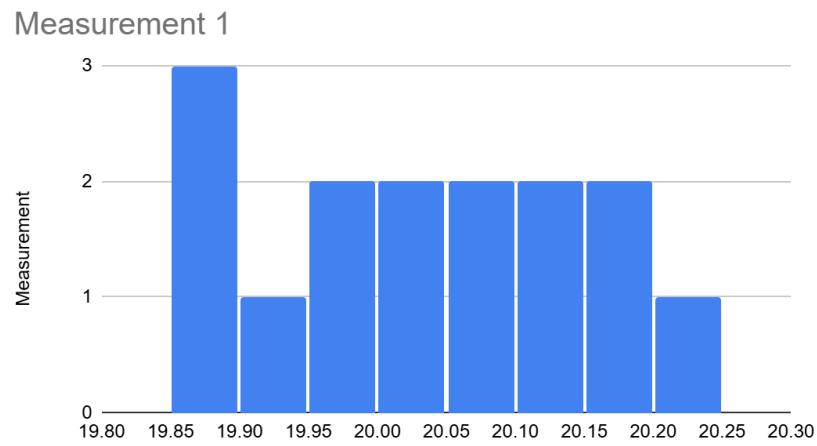


Figure 54. Histogram of Measurement 1

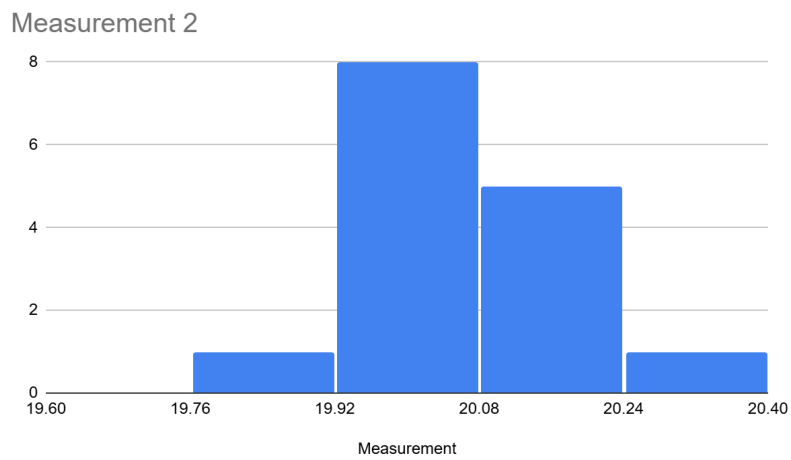


Figure 55. Histogram of Measurement 2

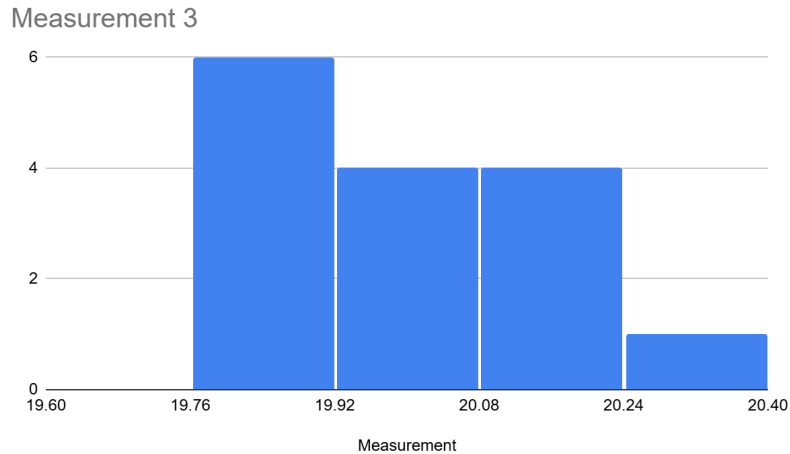


Figure 55. Histogram of Measurement 3

Materials in Prototype:

- Ender 3 Max
- Dremel Tool
- .25" Plywood
- 2 Metal Clamps
- Rubber bands

All of these parts have been clearly defined in the original analysis of the Ender 3. The only additional parts are MDF, which is manufactured as a solution of wood and clamped right now. Additionally, the clamps were brought from the Jackson Wood Shop, and placed on this part. I have an entire prototype and images attached below.

This prototype differs from my CAD design due to a more basic attachment for the Dremel tool. Also, the placeholder Dremel tool used in the CAD model had a slightly different shape – its handle is more curved and doesn't fit as well within the Ender 3's frame. In my engineering drawings, the "Wood Clamps" were homemade and 3D printed in PLA. The clamps on the physical prototype, however, are store-bought and were found in the Jackson Innovation Wood Shop.

I also did not 3D print an attachment for the Dremel tool, which turned out to be helpful since it fits

snugly within the bearings and small nozzle housing. I plan to iterate on this by designing a small 3D printed attachment to encompass that housing, which I initially thought wasn't reusable, and attach it later.

Another factor I hadn't accounted for is the G-code – I haven't yet been able to run the Ender CNC in its intended repurposed function. I will send video proof once I get it running.

The lab worksheet for Lab 7 is attached in the form of three tables and histograms above, in a more integrated way. I decided to measure the width of the 80/20 aluminum on the upper extrusion.

To update the CAD file, I would need to fully manufacture a new attachment. Other than that, the only difference is a lack of clamps used, opting for store-bought ones.

Here is an attached drawing of the C-Clamps in question:

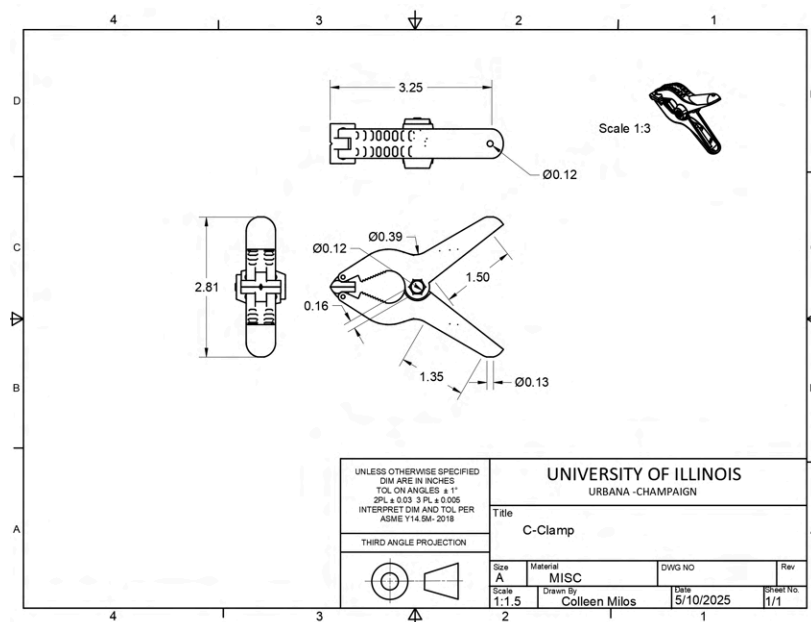


Figure 56. C-Clamp Drawing

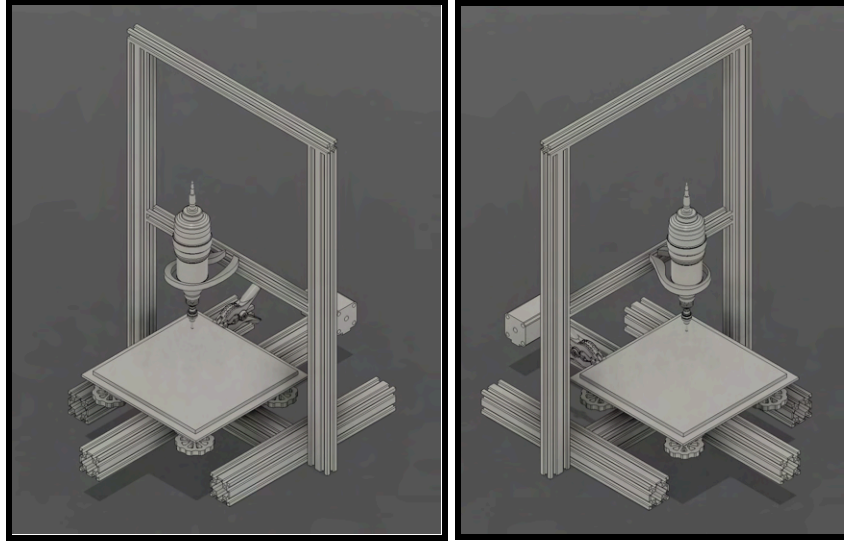


Figure 57. Updated CAD File

This is more updated and more realistic, matching the prototype I have currently.



Figure 58. Prototype View

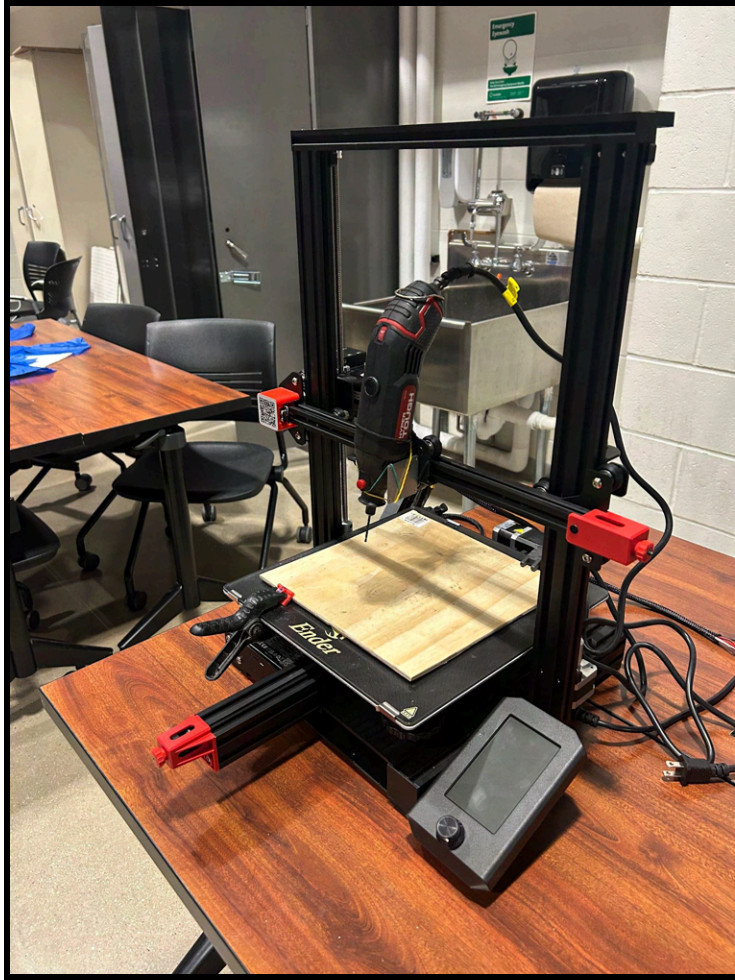


Figure 59. Prototype Additional View

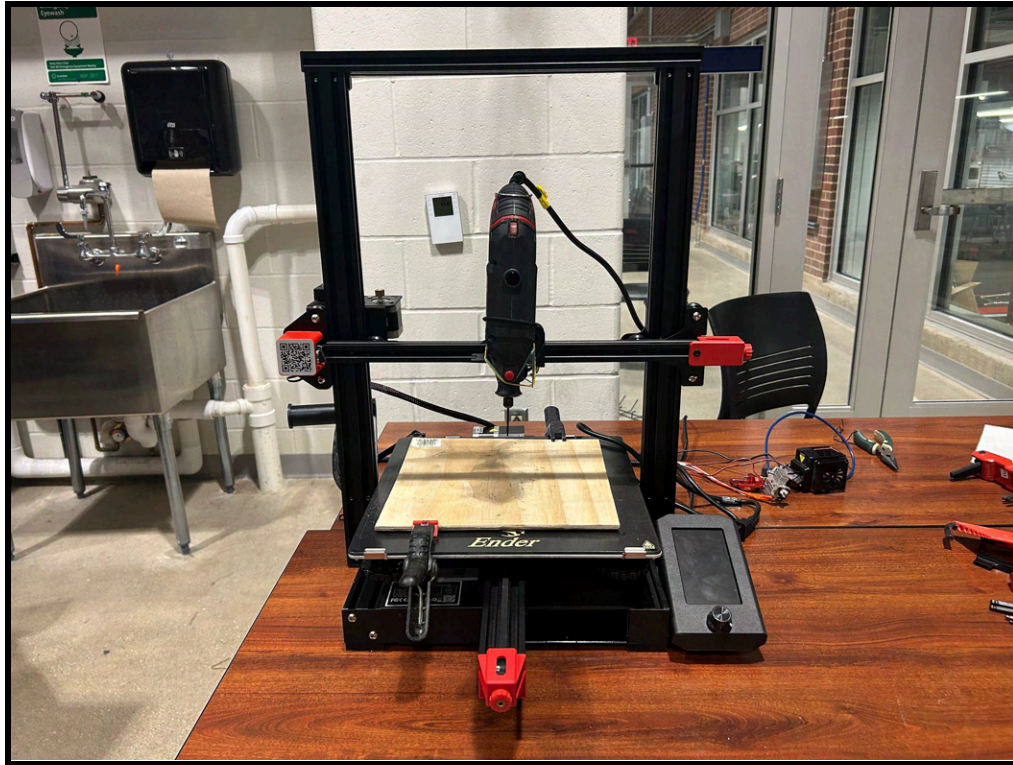


Figure 60. Prototype Front View

2.4.

Function	Essential Part
XYZ Motion	Frame and Motion Components
Tool Function	Dremel Tool
Work Holding	Clamps
Electronics	Power supply, stepper motors, motherboard
Support	Mounts, Brackets, MDF

The estimated minimum number of parts is 11, with a base frame, dremel, 3 stepper motors, a motherboard, power supply, GT2 belt system, mounting bracket, bed/work surface, clamp system, and harness.

The actual number of parts is more around 28, accounting for the frame, fasteners, 5 motors, full dremel, and cables, fans, brackets, and the PCB in the Ender 3.

The assembly time would be around 5 minutes given an already made Ender 3, with just the 3D attachment and all parts attainable. Setting up the MDF and clamping it is not incredibly time consuming, but calibration and software is far more time consuming.

Some observations are as noticed:

- A High part count due to reused Ender 3 components; could be reduced by consolidating 3D printed parts (e.g., brackets and mounts).
- Fasteners dominate assembly time; consider snap-fit or slot features in future versions.
- Tool and clamp integration is rudimentary but functional; a redesigned combined tool holder + clamp could reduce total parts.
- Wiring harnesses and motor mounts could be modularized to reduce individual part handling.
- It is difficult to arrange the different softwares at hand and integrate them, but in the world of prototyping, that is an issue for another time.

3. REFLECTION

Throughout this assignment, I got a much better understanding of how important material use and recovery is when it comes to design and manufacturing. One of the biggest things I learned was how to actually think about a material's full lifecycle—not just how strong or cheap it is, but how recyclable or reusable it is and how it affects the environment. When working with the Ender 3 Max, I noticed that so many parts—like the aluminum rails, screws, and motors—could be reused for something else without needing to be melted down or remanufactured. That made me lean toward reusing as many parts as I could instead of replacing them, which is why I stuck with things like the hotbed and frame instead of finding new components.

When I started Part 2, I thought turning a 3D printer into a CNC machine would mostly mean just adding a rotary tool and adjusting the code. But once I actually started building, I ran into a bunch of unexpected problems—like the motor being too weak for the weight, the frame not being as rigid as I thought, and even the shape of the Dremel not fitting like I designed in CAD. I didn't realize how tricky it would be to make good G-code either. All of that showed me how real prototyping is way more back-and-forth than I thought, and how every design choice kind of depends on how all the parts physically work together.

Looking back, I'm proud of how creative and thorough I was with coming up with ideas, trying out different sketches, and building an actual prototype that worked. I think I did a good job keeping things low-cost, reusing as much as possible, and keeping the project aligned with sustainability goals. If I could go back and do anything differently, I'd test the actual fit of the parts before finishing the CAD. I locked things in a little too early and ran into some fit issues because of it. Adding more flexibility to the model could have helped with that too.

After this whole process, I see how important it is to design not just for easy manufacturing but also for reuse. I'll definitely carry that mindset forward. I've also realized that prototyping and designing aren't just steps you do one after the other—they need to constantly feed back into each other if you want your final product to actually work.

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